St. Michael's University Hospital - Bone Marrow Transplant & Leukemia Service
INPATIENT DISCHARGE SUMMARY

ADMISSION DATE: December 26, 2023
DISCHARGE DATE: January 30, 2024

ADMITTING DIAGNOSIS: Newly Diagnosed Acute Myeloid Leukemia.

DISCHARGE DIAGNOSES:

1. Acute Myeloid Leukemia with t(9;11)(p21.3;q23.3), in Morphologic and Cytogenetic Remission.
2. Induction Chemotherapy-Related Febrile Neutropenia.
3. Severe Oral Mucositis, Resolved.
4. Profound Pancytopenia, Recovering.

HOSPITAL COURSE SUMMARY:
Ms. Brooks is a 48-year-old female with no significant past medical history who presented to an outside hospital on December 26, 2023, with a 2-week history of fever, chills, and a non-productive cough. She was found to have pancytopenia and was transferred to our facility for further management.

A bone marrow biopsy performed on admission confirmed the diagnosis of Acute Myeloid Leukemia with 85% blasts. Cytogenetic analysis revealed a **t(9;11)(p21.3;q23.3)** translocation, which involves the **KMT2A gene**. This places her in the **ELN 2022 Intermediate-risk** category. Given her young age, excellent fitness level (ECOG 0), and the specific genetic subtype of her leukemia, the decision was made to proceed with a more intensive induction regimen than standard 7+3.

After placement of a triple-lumen PICC line, she began induction chemotherapy on December 28, 2023, with the **FLAG-Ida** regimen:

- **Fludarabine:** 30 mg/m²/day IV on Days 1-5.
- **Cytarabine (Ara-C):** 2 g/m²/day IV on Days 1-5.
- **Idarubicin:** 12 mg/m²/day IV on Days 1-3.
- **G-CSF (Filgrastim):** Started on Day 6.

Her inpatient course was prolonged and marked by the expected toxicities of this intensive regimen:

- **Myelosuppression:** She experienced a profound and prolonged period of aplasia. Her Absolute Neutrophil Count (ANC) was <100 cells/uL for 20 consecutive days.
- **Febrile Neutropenia:** On Day +8 post-chemo, she developed a fever to 39.5°C. A full septic workup was initiated, including blood cultures, urinalysis, and chest CT. She was started on empiric broad-spectrum antibiotics (Piperacillin-Tazobactam and Vancomycin). Her blood cultures remained negative, and her fevers resolved after 5 days. She completed a 14-day course of antibiotics.
- **Mucositis:** She developed debilitating Grade 4 oral mucositis, which severely limited her ability to eat or drink. This required management with a Patient-Controlled Analgesia (PCA) pump with hydromorphone for pain control and Total Parenteral Nutrition (TPN) for nutritional support for a total of 12 days.

- **Transfusion Support:** She required extensive transfusion support throughout her nadir: 14 units of packed red blood cells and 22 units of apheresis platelets.

Response to Therapy:
Her peripheral blood counts began to show signs of recovery around Day +25. A Day +28 bone marrow biopsy was performed on January 25, 2024, to assess her response to induction. The results were excellent:

- **Morphology:** Revealed a recovering, normocellular marrow with <1% blasts.
- **Flow Cytometry:** No evidence of residual disease.
- **Cytogenetics:** FISH analysis for the KMT2A rearrangement was negative.
- **Conclusion:** She is in a Complete Remission.

Condition at Discharge:
Ms. Brooks has recovered remarkably. She is afebrile, hemodynamically stable, and her blood counts are safe for discharge (ANC 1.8 K/uL, Hgb 9.1 g/dL, Plt 85 K/uL). She is fully transfusion-independent. Her mucositis has resolved, and she is tolerating a regular diet. She is ambulatory and her pain is controlled with oral medication. Her PICC line remains in place.

DISCHARGE PLAN:

1. **Medications:**
    - **Levofloxacin** 500 mg PO daily (antibacterial prophylaxis).
    - **Posaconazole** 300 mg PO daily (antifungal prophylaxis).
    - **Acyclovir** 400 mg PO BID (antiviral prophylaxis).
    - **Oxycodone** 5 mg PO every 4-6 hours as needed for residual oral pain.
2. **Follow-Up:**
    - Return to Leukemia Clinic in 1 week (February 6, 2024) for CBC check and clinical assessment.
3. **Future Treatment Plan:**
    - **Bone Marrow Transplant Consultation:** She has an appointment with the BMT service next week. Given her intermediate-risk disease, an **allogeneic stem cell transplant** in first remission offers her the best chance for a long-term cure. We have initiated HLA-typing for her and her siblings.
    - **Consolidation:** The goal is to proceed directly to transplant. If a donor is not immediately available, she will require at least one cycle of consolidation chemotherapy (e.g., with HiDAC) as a bridge.

The patient and her family have been provided with extensive education on infection precautions, medication management, and when to call the 24/7 oncology line.

PATIENT: BROOKS, Judy
MRN: AML045
DOB: April 12, 1975